

Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference

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Motivating problem

Effect of healthy food access on obesity rates

Poor healthy food access been associated with:

- high systolic blood pressure (Suarez et al., 2015)
- hypertension and cardiovascular disease (Kelli et al., 2019)
- obesity (Thomsen et al., 2016)

Challenges include:

- **Interference:** cross-border shopping
- **Heterogeneous effects:** magnitude and direction of effects can vary by community characteristics

Research question: What is the effect of healthy food retailers on ZIP Code obesity rates and how does it vary for different ZIP Code characteristics?

Causal inference with interference

Interference

- **No interference assumption:** The treatment/exposure of one unit does not impact the outcome of another unit
- Often violated in studies with spatial treatments or exposures

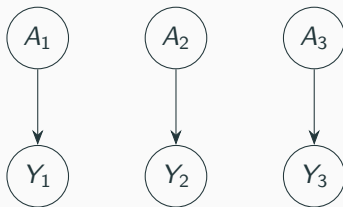


Figure 1: No interference

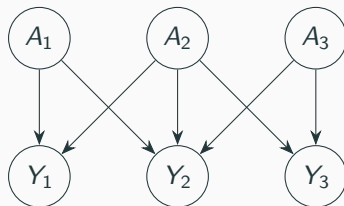


Figure 2: Spatial network interference

Notation and Potential outcomes

- $\mathbf{W}$ = adjacency matrix
 - w_{ij} = strength of connection between units i and j
 - if $w_{ij} = 0$, units i and j are unconnected
- $\mathcal{N}_i$ = set of units that are connected to unit i
 - Call this set the neighbors of unit i
- $(i, \mathcal{N}_i, \mathcal{N}_{-i})$ = partition of all units
- Potential outcome: $Y(A_i, \tilde{A}_i = f(\mathbf{A}_{\mathcal{N}_i}))$
 - $(A_i, \tilde{A}_i)$ = joint treatment
 - $f(\mathbf{A}_{\mathcal{N}_i})$ is a summarizing function of the neighboring exposures/treatments

- **Direct effect (DE):** effect of unit-level treatment (A) on Y holding neighbor-level treatment fixed

$$DE(\tilde{A} = \tilde{a}) = E[Y(1, \tilde{a}) - Y(0, \tilde{a})]$$

- **Indirect (spillover) effect (IE):** effect of neighbor-level treatment ($\tilde{A}$) holding unit-level treatment fixed

$$IE(A = a) = E[Y(a, 1) - Y(a, 0)]$$

- For simplicity, we consider binary A and $\tilde{A}$

Assumptions

1. Consistency: $Y_i = Y_i(\mathbf{A})$
2. **Neighborhood interference**: Given a function f , if $f(\mathbf{A}_{\mathcal{N}_i}) = f(\mathbf{A}'_{\mathcal{N}_i})$, then $Y(A_i, \mathbf{A}_{\mathcal{N}_i}) = Y(A_i, \mathbf{A}'_{\mathcal{N}_i})$ (Forastiere et al., 2021; Hudgens and Halloran, 2008)
3. Unconfoundedness of the joint treatment:
 $Y_i(a, \tilde{a}) \perp A_i, \tilde{A}_i | \mathbf{X} \quad \forall a, \tilde{a}, \quad \forall i$
4. Positivity: $f(a, \tilde{a} | \mathbf{X}) > 0 \quad \forall \mathbf{x} \quad \text{s.t.} \quad f_{\mathbf{X}}(\mathbf{x}) > 0$

Bayesian additive regression trees (BART)

- Ensemble machine learning method that models the outcome as a sum of weak learners
- Bayesian framework: places priors on tree structure and parameters and provides uncertainty quantification via credible intervals
- Often used for causal inference problems (Hill, 2011; Dorie et al., 2022)

Bayesian causal forests (BCF)

- Extension of BART designed for causal inference and heterogeneous treatment effects (Hahn et al., 2020)

$$Y_i = \mu(\mathbf{X}_i, \hat{\pi}(\mathbf{X}_i)) + \tau(\mathbf{X}_i)A_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- $\mu(\mathbf{X})$ = prognostic function
- $\tau(\mathbf{X})$ = moderating function
- Include estimate of the propensity score in the prognostic function

Joint treatment Bayesian causal forests (jtBCF)

- Extension of Bayesian causal forests for joint treatments
- Separate moderating functions for both A , $\tilde{A}$, and their interaction

$$Y_i = \mu(\mathbf{X}_i, \hat{\pi}_A, \hat{\pi}_{\tilde{A}}) + \tau_1(\mathbf{X}_i)A_i + \tau_2(\mathbf{X}_i)\tilde{A} + \tau_3(\mathbf{X}_i)A\tilde{A} + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

- $\mu(\mathbf{X})$: prognostic effect
- $\tau_1(\mathbf{X})$: direct effect (own treatment)
- $\tau_2(\mathbf{X})$: indirect (spillover) effect
- $\tau_3(\mathbf{X})$: interaction effect

- We use the SoftBart R package (Linero, 2022) to implement our model
 - SoftBart R package also includes functionality to develop custom BART models
- Soft BART replaces binary splits with smooth gating functions (e.g., logistic functions) so each observation can partially belong to multiple leaves (Linero and Yang, 2018)

Simulations

Simulation setup

- Use Texas ZIP Codes as set of units ($n = 1927$)
- Simulate three unit-level covariates and compute three neighborhood-level covariates, which are neighborhood means of the unit-level covariates
- Consider linear/nonlinear $\mu(\mathbf{X})$
- Consider homogeneous/heterogeneous $\tau_1(X) - \tau_3(X)$
- Treatment:

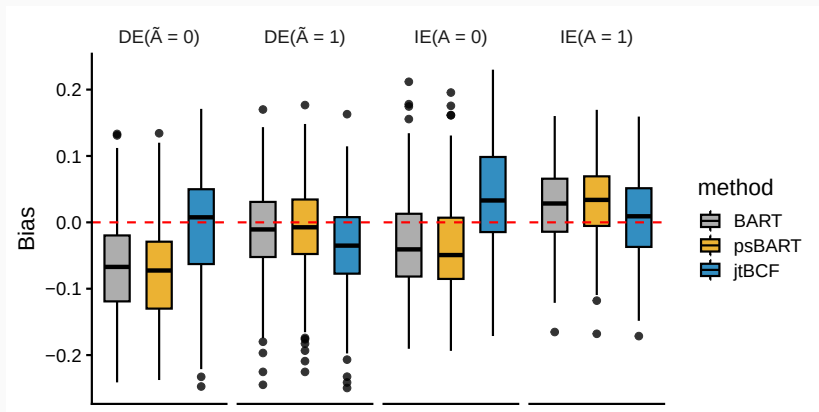
$$A \sim 0.8 \Phi(-1.25 + \mu(X)/\text{sd}(\mu(X)) + 0.5X_5) + u_i/10$$

$$\tilde{A}_i = \mathbf{1}(\bar{A}_{\mathcal{N}_i} > 0.5)$$

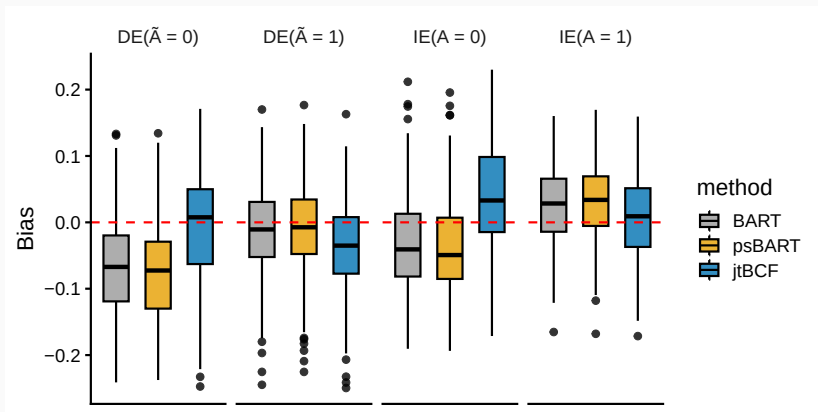
- Outcome:

$$y \sim N(\mu(X) + \tau_1(X)A + \tau_2(X)\tilde{A} + \tau_3(X)A\tilde{A}, 1)$$

Simulation results (Linear/Homogeneous case)



Simulation results (Linear/Heterogeneous case)



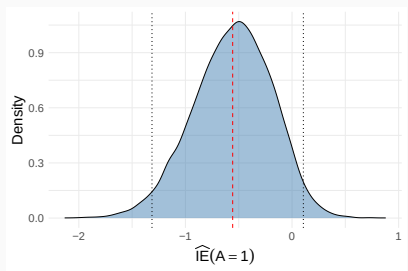
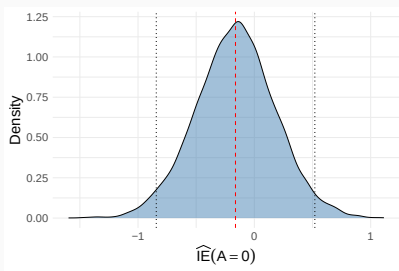
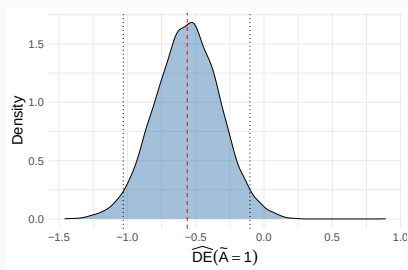
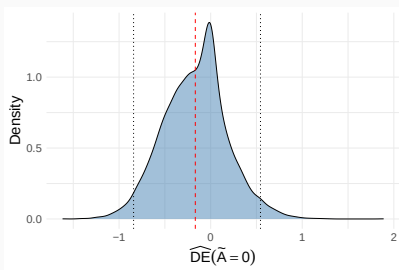
Coverage - Direct and Indirect effects

Setting	Estimand	Soft BART	Soft psBART	Soft jtBCF
Homo				
	$DE(\tilde{A} = 0)$	0.80	0.78	0.94
	$DE(\tilde{A} = 1)$	0.94	0.94	0.93
	$IE(A = 0)$	0.90	0.86	0.93
	$IE(A = 1)$	0.91	0.90	0.96
Hetero				
	$DE(\tilde{A} = 0)$	0.82	0.82	0.96
	$DE(\tilde{A} = 1)$	0.92	0.93	0.93
	$IE(A = 0)$	0.93	0.93	0.92
	$IE(A = 1)$	0.95	0.95	0.97

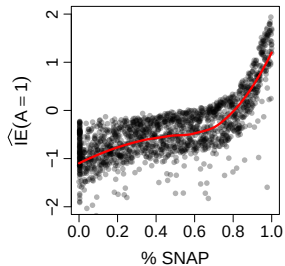
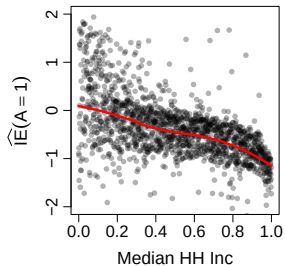
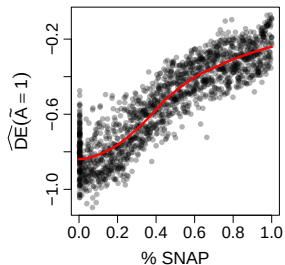
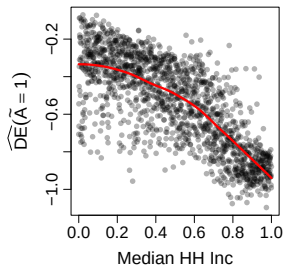
Case study: Effect of healthy food access on obesity rates in Texas

- **Unit of analysis:** Texas ZIP Codes in 2022
- **Outcome:** Obesity prevalence ($y \in (0, 100)$)
 - Model-based estimate adjusted for age, sex, race, and education
- **Exposures:**
 - A : Indicator for ≥ 1 healthy food retailer in the ZIP Code.
 - $\tilde{A}$: Indicator for ≥ 1 healthy food retailer in neighboring ZIP Codes.
- **Covariates:** Median household income, % SNAP, unemployment, % commuting by public transit, NDVI, % married, % veterans, and % with disability
- **Neighborhood:** Informed by driving times between ZIP Code centroids

Posterior distributions of direct/indirect effects



Results: Heterogeneous effects



Discussion

Conclusions

- jtBCF model offers bias reduction in estimating direct and indirect effects under interference compared to BART and psBART
- Provides strong coverage of direct and indirect effects
- Limitations:
 - Model has increased variability due to noise coming from three separate moderating functions
 - Motivating example has some weaknesses:
 - Data is ecological
 - Obesity rate at ZIP Code level is model-based estimate (takes into account race, sex, age, and education)
 - Food retailer list is not exhaustive; does not include farmer's markets or food banks

Thank you!



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Any questions?

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$$\mu(\mathbf{x}) = \begin{cases} 1 + 0.7X_1 - 0.6X_2 + X_3 + 0.3X_4, & \text{linear} \\ 0.5 + 0.6|X_1 + 1| - 0.8\sin(\pi X_2) + X_3 - 0.6X_4 + 0.3X_1X_4, & \text{nonlinear} \end{cases}$$

$$\tau(\mathbf{x}) = \begin{cases} 0.4A + 0.4\tilde{A} + 0.2A\tilde{A}, & \text{homo} \\ 0.3(1 + \frac{X_1}{2} + X_3)A + 0.3(1 + \frac{X_2}{2} + X_3)\tilde{A} + 0.15(X_3 + X_6)A\tilde{A}, & \text{hetero} \end{cases}$$

Results: Overall effects

- $DE(\tilde{A} = 0) = -0.17(-0.84, 0.54)$
- $DE(\tilde{A} = 1) = -0.56(-1.03, -0.10)$
- $IE(A = 0) = -0.16(-0.86, 0.50)$
- $IE(A = 1) = -0.56(-1.25, 0.17)$

Isolated nodes

- Since we use driving times to inform the neighborhood adjacency, some ZIP Codes are isolated (no neighbors)
- Often it is assumed $\tilde{A} = 0$ for isolated nodes
- We drop these ZIP Codes from the analysis since it has been shown that estimation of spillover effects is biased when we assume $\tilde{A} = 0$ for isolated nodes (Kim, 2025)